Unit - 4

Cloud Security

Security is core concerned because an organization or consumers trust for their critical information on cloud.

which spread all over the world and not direct control of any organization or consumer.

Cloud Security is the protection of data stored online from theft, leakage, deletion.

Methods of providing cloud security include firewalls, Penetration testing, VPN's.

The main Goal of cloud security is to provide the cloud platform application & user data as well as to provide an environment without worry for consumer by managing all security issue by the cloud providers.

Different Main Concern of Cloud Security :-

Third Party Handling :- This data is handled by a third party so they cannot give 100% security.

Cyber attack :- Different Type of attack are there and there is very challenging issue in cloud security

Insider threats :- Employee can use their authorized access to an organization cloud based service to misuse.

Legal Liability :- In this way court case file against you & against the cloud company or cloud vendor's.

Lack of Support :- How much support company provide to you, because prices are not fixed.

Lack of Standard :- Different cloud procedures may not follows the same standard.

Threats in cloud Security :-

DOS :- Denial of Service attack.
In this attaker tries to goes server down, so speed and performance goes down.

MITM :- Man in middle attack
The main motive of this attack is to theft important data b/w two communicable parties.

NS :- Network Sniffing
In this attaker can monitoring whole traffic of network.

PS :- Port Scanning
In this attacker easily know about port no, which port is used by the user.

SIA :- SQL injection attaker
In this hacker or attacker tries to steal confidential information from database, this attack is done by SQL queries.

XSS :- Cross Side Scripting attack
In this attacker can leave the link on the website, than user can give her | his detail on the website.

Cloud Security Fundamental :-

Confidentiality It is keeping the information private, it prevent the unauthorized access of any other. In this information kept secure, confidential

Integrity :- In this modification of data is not allowed by any unauthorized person, data is consistant in any mode.

Availability :- It means that information is accessible by authorized user.

Cloud Security Services :-

Different cloud services are there :-

Authentication :- Authentication is the process of verifying who a user, it is done before authorization.

Authorization :- while authorization is the process of verifying what they have access to or validated, this process is done ~~before~~ After authentication.

Auditing :- It is a process of audit all the information, whole process which is used in security services.

Accountability :- It provides a way forward in dealing with data protection issue arising from handling personal data in cloud.

Cloud computing Security Challanges :-

- if we migrate into cloud so cost will be normal for the cloud consumer.

- Availability :- Accessibility of services are reliable means services access easily & fast.
- The down time should be very less in any case of failure of cloud providers.
- Lock In :- if once a information migrate into cloud than this information should be isolated by others.
- Data Breaches :- In any case data can't be break or steal.

Legal issues in cloud computing :-

- Physical location of your data :-

Physical location not shared by user, user don't know the location of data.

- Security of cloud vendors :-
  - Hack data center (what will do)
  - Disaster (who will be responsible)

- Service level agreement :- Signed on legal paper
  - Down time,
  - Process of failure.

Lack of laws dedicated to Cloud Computing :

→ Some of the companies don't have any rule or legal paper related to data.

Cloud Security Implementation Policy :-

Organization → committee → Management →
Security rules → set of rule of security Policy

→ A Policy is an essential and fundamentals element of Security Policies. A Security Policy must be implemented after creating the foundation of Policies, Standard, guidelines, Procedures.

Different Policies are there :-

Regulatory Policy
Advisory Policy
User Policy

Regulatory Policy :- This Policy is very detailed and Specific to the industry and organization. These organization may be financial company company and public utility.

Advisory Policy :- This Policy is not mandatory but strongly suggested and helpful to ensure Security of System.

User Policy :- This Policy is mandatory for the user who are associated with any organization or company, because any wanted user can't break the any data of organization or company.

Cloud Computing Security design Principle :-

Principle of least Privilege :-
An important concept in computer security, is the practice of limiting access right for users to the bare minimum permissions they need to perform their work.

Seperation of duties :- each user of organization complete their task independently.

Data in transmit protection :-
when data in transportation or on process one place to another place data should remain constant.

Seperation b/w users :- Seperation is compulsory b/w users, because each task & each user complete a task uniquely so each are isolated from each other.

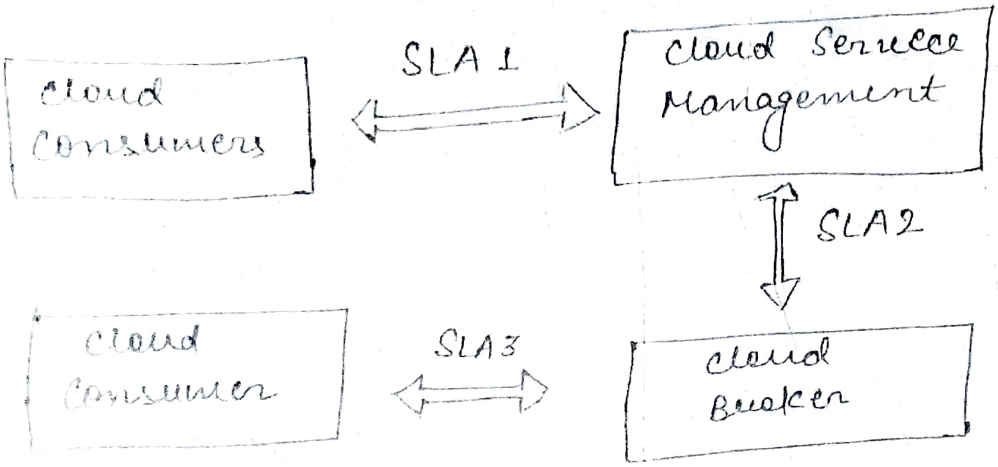
Government Framework :- Government design different set of rules and law to avoid the risk.

Audit Record Maintenance :- If any problem is occur this audit record is responsible for solve these problem

SLA (Service Level Agreement) :-

A cloud SLA is an agreement between a cloud service provider and a customer that ensures a minimum level of service is maintained.

A service-level agreement is a commitment b/w a service provider and a client. particular aspects of the service - quality, availability, responsibilities.



Always two parties are include in SLA.

Cloud Providers :- This provide the legal agreement because security is handled by the providers.

Cloud Consumer :- Consumer always take the SLA because if any provider are doing cheating then this SLA help to the user all the rules are in the society.

Trust management :-

- Computing is the growing technology a System Proposed that implementation of Trust management System for cloud computing.
- A user trust a cloud service with respect to performance, security and privacy based on the identity of the providers.
- latest model is a collection of rules that inform application on how to decide them.
- Trust Management is an issue that need special attention and it is an important concept of cloud security.
- Trust Management in cloud computing involved three aspect :-
  - Data in privacy protection and data integrity
  - Trusted cloud computing over data
  - Centre security awareness
  - cloud architecture trusted navigation
- Various techniques used for trust management
  1. Data coloring & watermarking.
  2. Encryption & Cryptography.

Data Security in cloud computing :-

In this we will make an understanding of data security disaster recovery business continuity and risk mitigation.

Business continuity and disaster recovery.

Business Continuity Planning (BCP) and Disaster recovery Planning (DRP) include in cloud security.

These important business process are independant on cloud based application software robustness and security.

Disaster :- disaster is a rapidally occuring and unstoppable even that cause of loss of life or damage to system. It is serious description occuring over a short or long period of time.

DRP :- In this action should be taken before, after or during the disaster, huge volume of data loss due to disaster.

BCP :- In this design to keep a business running reduce the risk of financial loss and to evaluate a company's capability to recover from disaster.

Back-up recovery :- Back up services are the main element of the disaster recovery plan.

- Agreement on the data center.
- Subscription services among the third party.
- Multiple data center are used.

Risk Mitigation :- Risk Mitigation is a Systematic approach to reduce the exposure of risk & Possibility of it's occurence.

It is the identification, evaluation, Prioritization of risk followed by economical application of resources to minimize, monitor, Control the Possibility or maximize the realization of opportunities.

There are 5 Types of Risk Mitigation :-

Identify the Risk :- You and your team uncover recognize and describe the risk.

Analyse the Risk :- If risk is identified that we analyze the risk & work on it.

Evaluate the Risk :- In this we Evaluate the risk than we take the decision risk is accepatable or not then we start the treatment accordingly.

Treat the Risk :- This is also known as risk response planning and we treat the risk easily in this step.

Monitor & Review the risk :-

In this we monitor the risk and review and track the whole risks of our system.